SOCIOQUEST: AN AI-DRIVEN BEHAVIORAL SIMULATION AND CYBER-RESILIENCE FRAMEWORK FOR SOCIAL ENGINEERING DEFENSE

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CARL ANTHONY M. ESTURAS
GIAN J. VILLARENTE
WILL MAVERICK C. VERGARA

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CHAPTER 1

INTRODUCTION

Background of Study

The modern digital landscape has shifted the primary target of cyberattacks from software vulnerabilities to the human element. While traditional security measures like firewalls and encryption have become highly sophisticated, the "human layer" remains the most accessible entry point for malicious actors. Social engineering tactics have evolved beyond simple, easily detectable scams into complex, psychologically driven deceptions. Recent empirical studies emphasize that the exploitation of human vulnerabilities, driven by cognitive biases and a lack of contextual awareness, is the primary catalyst for successful cyber intrusions [1]. Standard security awareness training is frequently too static and disconnected from reality to be effective, as it fails to prepare individuals for the pressure of a real-time attack delivered through everyday communication channels. Consequently, researchers advocate for adaptive, simulation-based interventions over traditional compliance-based training, demonstrating that experiential learning significantly reduces susceptibility to advanced phishing and manipulation techniques [2].

Within the Philippine context, the rapid shift towards digital transformation and e-commerce has been accompanied by an escalating frequency of cyber threats targeting the workforce [3]. Threat actors actively leverage trust, urgency, and local familiarity to trick individuals into compromising sensitive organizational data. Investigations into recent high-profile breaches across the country reveal a critical knowledge gap among non-technical professionals, demonstrating that current preventive measures and standard awareness campaigns are insufficient against evolving deceptions [3]. Gamified and interactive educational frameworks are being increasingly recognized as necessary tools to actively engage users and improve information retention regarding social engineering threats [4]. By moving away from passive learning models, these interactive approaches help build a proactive defense mechanism among employees, reinforcing the necessity for experiential cybersecurity training in local institutions.

SocioQuest aligns with the Sustainable Development Goals (SDGs) prioritized by Davao del Norte State College (DNSC) by addressing the intersection of technology,

education, and institutional security. By implementing an AI-driven behavioral simulation, the project directly supports SDG 4: Quality Education through the creation of innovative, "future-ready" learning solutions that move beyond traditional, static training to build practical cyber-resilience. The framework also contributes to SDG 8: Decent Work and Economic Growth and SDG 9: Industry, Innovation, and Infrastructure by protecting Panabo City's growing administrative and commercial sectors from the disruptive economic impacts of social engineering attacks, effectively turning employees into a proactive "Human Firewall". Furthermore, the study upholds SDG 16: Peace, Justice, and Strong Institutions by promoting good governance and transparency through a structured reporting and reasoning process that safeguards sensitive organizational data from deceptive threats.

The study is specifically situated within the organizational environment of Panabo City, Davao del Norte. This locale is chosen because it represents a growing hub of both public administration and private commerce where digital transformation is rapidly increasing the surface area for potential cyber threats. In the neighboring area of Davao City, assessments of cybercrime victimization highlight significant vulnerabilities among the population, particularly concerning risky online behavior and exposure to digital fraud [5]. Furthermore, research evaluating the cybersecurity behavior of individuals in the Davao region underscores that perceived behavioral control and hands-on awareness are strong predictors of secure digital practices [6]. SocioQuest is designed to bridge the gap between theoretical knowledge and practical cyber-resilience by introducing an AI-driven behavioral simulation. The project moves away from traditional classroom-style learning and implements a live "Office Siege" environment where the stakes are simulated but the experience is immersive. By using artificial intelligence to generate authentic-sounding messages in Bisaya and Taglish, the framework forces users to develop a critical eye for subtle fraud indicators within the safety of a controlled game. Ultimately, this localized approach not only identifies who is vulnerable but also teaches individuals how to deconstruct an attack, turning every employee into a proactive "Human Firewall".

The study is situated within the organizational landscape of Panabo City, Davao del Norte, a strategic locale characterized by its rapid digital transformation as a hub for public administration and private commerce. This growth has expanded the "attack surface" for local institutions, making them prime targets for malicious actors. The proponents observe that Panabo's local organizations face a critical gap in

cyber-resilience because traditional security training remains too static and disconnected from the high-pressure reality of real-time attacks. Consequently, employees, the "human layer", remain the most accessible entry point for cyber threats, leaving organizational data highly vulnerable to sophisticated deceptions.

To address the evolving landscape of cyber threats, the proponents have identified specific problems aligned with the project's objectives. First, the rise of localized deceptions necessitates an AI-Driven Multi-Lingual Attack Engine to simulate threats in English, Taglish, and Bisaya, addressing the lack of localized detection. Second, the invisibility of threats is solved by a Multi-Tenant Siege Map Dashboard, providing real-time tracking of department-level security health across independent organizations. Third, the proponents address the lack of structured reporting by implementing a Fraud Factor Analysis Module, where users pinpoint specific "Glitch Markers" to validate their reasoning.

Furthermore, the project replaces passive learning with an immersive "Defend-Analyze-Repair" Loop. In this gamified cycle, users Defend by identifying attacks on their real devices and verifying them via a Dual-Reality Dashboard; Analyze the threat through the Fraud Factor module; and Repair their 2D Department Base on a strategic map using earned resilience points. To ensure safety, the framework features an Admin-Controlled Attack Scheduler, allowing IT administrators to precisely time simulations. To prevent misuse, every simulated communication includes a Universal Safety Indicator, clearly labeling the message as a "SocioQuest Simulation". Finally, a Link-to-Lecture Remediation framework ensures that when a user fails, they are immediately redirected to a targeted 2D lecture and recovery quiz.

Objectives of the Study

The general objective of this study is to develop SocioQuest, an AI-driven behavioral simulation and cyber-resilience framework designed to strengthen social engineering defense through gamified real-world attack simulations and localized threat detection. Specifically, the study aims to achieve the following:

1. To develop a multi-lingual attack engine capable of generating localized phishing emails, SMS, and voice call scripts in English, Taglish, and Bisaya utilizing generative AI to mirror modern, hyper-personalized threats.

2. To design an Administrative Control Panel that allows authorized administrators to manage, schedule, and trigger simulations during specific off-hours windows, while monitoring department-level resilience through threat heatmaps.
3. To create a 2D "Siege Map" Interface using web-native game engines to visualize department-level security health and resilience scores in real-time, allowing users to interact with their organization as a series of strategic bases.
4. To implement a web portal that allows users to report and verify suspicious communications and validate their reasoning through a fraud factor analysis module.
5. To establish a gamified progression system where users earn resilience points based to repair departmental bases and progress through dynamic difficulty levels.
6. To develop a Targeted Remedial Framework that utilizes a lecture mechanic, redirecting users to immediate, data-driven lectures and recovery quizzes following a simulation failure to ensure continuous improvement.

Significance of Study

SocioQuest highlights its transformative impact on the digital security landscape of local institutions, shifting from traditional reactive training to a proactive, AI-driven defense. By bridging the gap between theoretical knowledge and practical cyber-resilience, the project strengthens institutional security while contributing to the growing field of behavioral cybersecurity. The following are the study's main beneficiaries:

- **Local Institutions.** By implementing this framework, local organizations in Panabo City can significantly strengthen their security posture and transform their workforce into a proactive "Human Firewall". The system enables these institutions to minimize successful social engineering attacks, protect sensitive organizational data, and foster a more resilient security culture
- **IT Administrators.** The web-based "Siege Map" dashboard provides administrators with real-time, data-driven insights into department-level security health and resilience scores. This automates the identification of high-risk areas, allowing for more efficient resource allocation and targeted security interventions.
- **Employees and End-Users.** The gamified simulation environment offers an immersive learning experience that prepares individuals for the high-pressure

reality of real-time attacks. Users develop a critical eye for subtle fraud indicators and learn to apply structured reasoning to identify emotional triggers and sender anomalies in a safe, controlled setting.

- **Community and Local Governance.** The study supports local digital transformation by ensuring that the growth of public administration and commerce in Panabo City is supported by a cyber-resilient workforce. This protects the community from the disruptive effects of financial and data-driven fraud.
- **Future Researchers and Developers.** This project establishes a foundation for future research on integrating generative AI and localized dialect-tuned content, specifically in English, Taglish, and Bisaya, into cybersecurity training frameworks. It lays the groundwork for more sophisticated, culturally relevant AI-driven behavioral simulations in educational and professional institutions.

Scope and Limitation

The scope of this project focuses on the development of a Multi-Tenant Web-Based Portal utilizing the FastAPI framework for the backend and React.js for the frontend to ensure universal access across all desktop and mobile browsers. The system employs a PostgreSQL relational database with a multi-tenant architecture to maintain strict data isolation between different organizations. The framework features an Automated Simulation Engine that orchestrates mock social engineering attacks via Email, SMS, and Voice Calls (Vishing). The timing of these attacks is governed by an Administrative Control Panel using the APScheduler library, which allows administrators to manually trigger simulations or enable Adaptive AI Scheduling based on department-level vulnerability profiles.

The technical core of the system is the AI-Driven Content Generator powered by the OpenAI GPT-4 API, which is programmed to produce authentic messages in English, Taglish, and Bisaya, while utilizing text-to-speech services for voice simulations. To maintain security, every communication is integrated with a Mandatory Safety Tag and a Cryptographic Watermark verified via the web portal. The project includes a 2D Strategic Map rendered through the Phaser.js game engine, visualizing organizations as departmental nodes with dynamic health bars and Predictive Threat Heatmaps. Users interact through a Behavioral Reasoning Survey that validates identification of seven

threat factors, followed by Psychological Feedback Loops. In instances of simulation failure, the system utilizes a Link-to-Lecture Remediation framework, where the FastAPI backend manages unique tracking URLs that redirect users to a targeted React-based lecture interface.

The study is geographically and functionally limited to the personnel of selected organizations within Panabo City, Davao del Norte. SocioQuest is strictly a simulation and training framework; it does not perform actual hacking, credential harvesting, or the distribution of live malware. The project boundaries exclude physical social engineering tests, such as tailgating or unauthorized office entry, and the system does not function as an automated mitigation tool. It will not integrate with or modify an organization's existing firewalls or email servers, and all simulations require manual administrator triggers or approved adaptive scheduling to ensure institutional safety. Additionally, simulation frequency and volume are strictly constrained by the API credit quotas of the integrated Twilio (SMS/Voice), SendGrid (Email), and OpenAI services. This framework is intended solely for training and research purposes and is not a replacement for comprehensive, enterprise-grade technical security suites.

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The "human layer" remains the most accessible entry point for cyber threats in Panabo City, as traditional security training remains too static to be effective. The Problem Tree analysis reveals that localized deceptions in regional dialects like Bisaya and Taglish often bypass standard awareness campaigns, while the invisibility of organizational security makes it difficult for administrators to track risks across departments. Because current training models are passive, knowledge retention is low, and employees lack a structured reasoning process to identify psychological deceptions. This leaves local institutions in a vulnerable state where employees are easily tricked by real-time social engineering.

SocioQuest transforms this reactive state into a proactive defense through an immersive behavioral simulation framework. Following the Objective Tree, the system utilizes a Multi-Lingual Simulation Engine to generate authentic, localized threats that prepare users for high-pressure reality. By implementing a Defend-Analyze-Repair loop, users engage in a gamified environment where they must identify fraud factors to earn resilience points and "repair" their departmental bases. The framework includes a Resilience Visualization Dashboard for real-time tracking and a targeted remedial framework that triggers immediate learning sessions after failures. This approach turns the workforce into a proactive "Human Firewall," securing Panabo’s digital transformation.

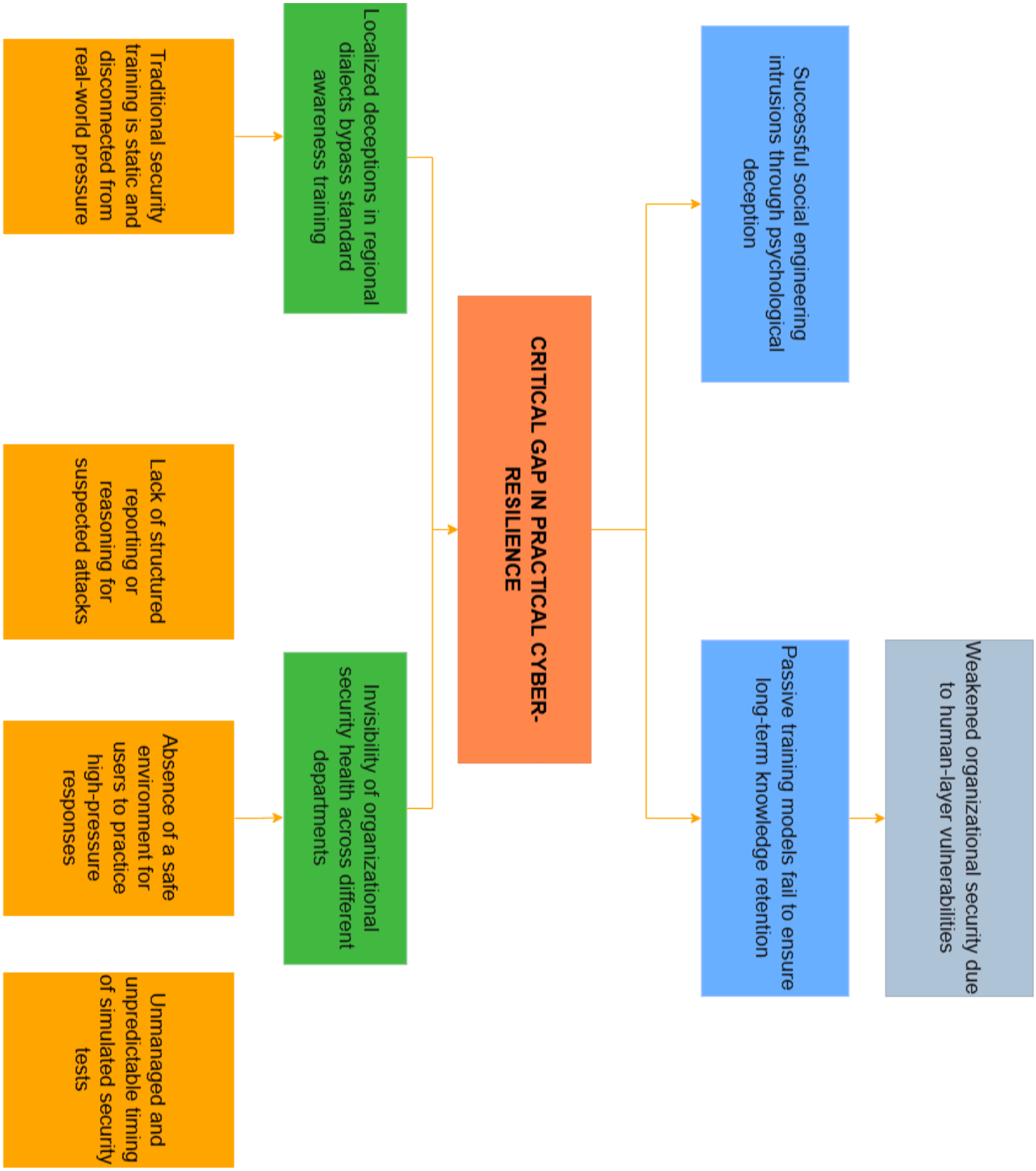

Figure 1. Problem Tree Analysis Diagram

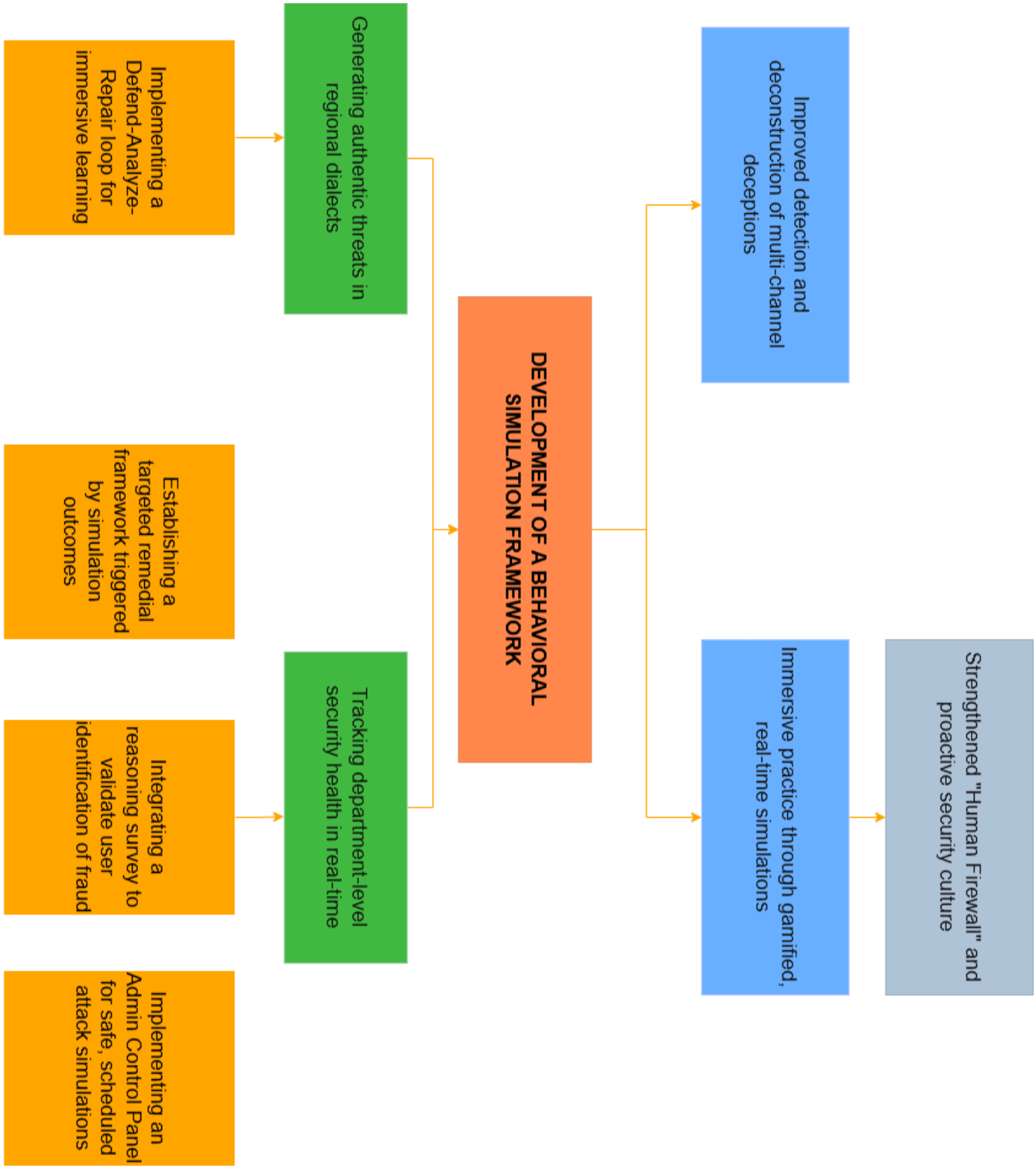

Figure 2. Objective Tree Analysis Diagram

Features	Natonton Framework	Cyber Battle Game	AIU Empirical Sim.	SocioQuest
Real-World Attack Delivery			✔	✔
Gamified Narrative & Progression	✔	✔		✔
Generative AI Attack Creation				✔
Department Base Repair Mechanics				✔
Dual-Reality Verification				✔

Table 1. Comparative Table